

The Cloud: The impact of digital infrastructure on the physical world

How the development of information and communication technologies have sparked new sectors, altering how energy is consumed globally, and how this growth can become more sustainable.

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December 2nd, 2024

Executive Summary

In modern society, there have been several emerging benefactors of information and communication technologies (ICT), including artificial intelligence, smart energy systems, manufacturing systems, self-driving cars, and more. In the US, nearly 5% of all energy consumption goes to computing power (Chen, 2023). To account for this growth, the ICT sector has explored its options to reduce its carbon footprint and energy usage. Of these options, most notably are workload management frameworks (WMFs), measuring energy consumption of software, and utilizing virtual technologies. WMFs allow modular data centers to optimize their geographical location and the dynamic nature of energy supply (particularly with implementing renewable energy sources). One such framework, SkyBox, has seen up to 46% in carbon footprint reductions, with the additive benefit of lowering cost and energy consumption (Sun et al., 2024). In terms of measuring energy consumption of software, there aren't many accessible ways to do this, but there are ways. By finding this metric of their applications, software developers will be able to work towards building "green software" and optimize the energy consumption of their programs. Virtual technologies, like server virtualization, allows for virtual machines to run on the same server, giving data centers the advantage of efficient scaling of resources when demand is changing. Server virtualization, in combination with increased server efficiencies, has facilitated a sixfold increase in compute instances with only a 25% increase in global server energy use (Masanet et al., 2020). Transitioning the ICT sector to the cloud has shown to consume fewer resources and electricity. Microsoft has even found that cloud services are 98% more carbon efficient than on-premises solutions (*"Energy Act of 2020 IT Sustainability Best Practice Guide"*). Despite all this, there is still more to get done. Policy must be pushed on these ICT companies in order to ensure a sustainable future for the sector, as well as the planet. Some of these possible policies include tracking usage effectiveness, implementing sustainable practices, and acquiring hardware that follow sustainability guidelines. It is important to consider the difficulties of implementing some of these suggestions, particularly when considering the complexities within the industry such as scalability, compatibility, and reliability.

Table of Contents

Executive Summary.....	1
Table of Contents.....	2
Introduction.....	3
Background.....	3
Discussion.....	4
Future Actions.....	6
References.....	7

Introduction

The many advantages available within day-to-day life have become abundant with a newly digitalized world—from a decline in commuting, being able to shop online, the rise of working from home, and even elearning—these changes may be causing a decline in energy use within other sectors. The accelerated development of information and communication technology (ICT) as a result of data-driven industries has shifted global energy consumption dramatically. The ICT sector is representative of most computer science work and energy consumption, and as of 2020, the industry accounted for 7% of global electricity consumption—a high mark compared to the estimated consumption of 1–1.6% in 2007 (Nouredine, 2024). Measuring and understanding the consumption of energy within the ICT industry and data centers has been difficult, though, with little international push towards the publishing of this information.

Background

Fortunately, there has been some push towards a greener and more sustainable ICT sector, with the US government tasking a council to recommend various sustainability practices for IT products and services. One of the most critical suggestions is moving to the cloud, as it has a direct correlation to sustainability and reducing carbon footprint (*“Energy Act of 2020 IT Sustainability Best Practice Guide”*). Some of the leading tech corporations who have done this have seen remarkable improvements; Amazon Web Services reported an 88% carbon footprint reduction after migrating to cloud services; and Google has seen a reduction in direct energy consumption for servers and cooling systems by 70–90% (*“Energy Act of 2020 IT Sustainability Best Practice Guide”*). Another similar option is using virtual technologies to enable more sustainable outcomes, such as server virtualization. This practice allows for multiple virtual machines to run on the same physical server, resulting in efficient scaling of resources to meet demand changes. This includes an emphasis on software developers implementing more efficient coding practices (“green software”) to reduce energy usage in computing instances, though precisely monitoring software energy use can be quite a challenge. Another ICT sustainability practice, parallelization, is the placement of multiple computing cores on a chipset. This allows for computations to be split up into smaller sub-tasks which can be performed simultaneously,

ultimately “yield[ing] better efficiency than simply improving clock frequencies” (Gelenbe & Caseau, 2015) on a chipset. Edge computing is another option in which a data center physically moves portions of its infrastructure significantly closer to its applications and services, which in turn results in more efficient computing jobs.

Tracking usage effectiveness and other data measures was another crucial recommendation provided by the tasked group. One particular way in which energy efficiency of data centers has been measured is power usage effectiveness (PUE). This metric is the ratio of total energy used by a data center to total energy consumed for computing operations. Though not mandatory to report this measure, more and more major data centers are reporting their PUE. There are other important metrics to be considered, such as water usage effectiveness (WUE) and carbon usage effectiveness (CUE), which similarly to PUE represent the ratio of total water or carbon used by data centers to the amount of water/carbon used for IT equipment. Note that none of these metrics should be prioritized over the other, as it can hinder sustainable outcomes; rather, data centers should look to balance all usage effectiveness ratings.

The prioritization of making ICT more sustainable will only continue to grow as more people gain access to ICT devices. A vivid illustration of this growth was the noticeable increase in the number of mobile phones in the world, which was more than 6 billion by 2013. To adjust for its growth, the ICT industry has innovated with notable efficiency gains, which have allowed ICT services to substantially minimize growth in energy use. While data center compute instances increased by over 550% globally, only a 6% increase was seen in electricity consumption from 2010 to 2018 (Masanet et al., 2020), and computations per kWh doubled about every year and a half from 2007 to 2012 (Gelenbe & Caseau, 2015). The potential for energy reductions in modern technology has largely been explored, but there still is a significant amount of room for improvement, especially with newer, more innovative technologies.

Discussion

Moving much of today’s data centers to the cloud has sparked another shift in the ICT sector. Now that most data centers are modular, there comes the challenge of transitioning to renewable powered modular data centers (rMDCs). The main issue of using renewable sources of energy to power data centers is the variability that typically comes with those sources. This

implies an unpredictable amount of energy in powering computing systems at modular data centers (Sun et al., 2024), which often results in degraded application performance as an aftereffect from task evictions and migrations. To optimize for the dynamic nature of the volatility patterns, workload management frameworks (WMFs) have allowed for significant improvements in the use of renewable sources of energy and environmental footprint. One such framework, representative of the former, is the SkyBox design model, which selects renewable energy farms as good candidates for placing modular data centers. It then selects complementary farms that have stable power output (power produced across different renewable energy sources and is stable for at least one week), which allows to better mask the unpredictable nature of renewable energy farms. This design also assists in ensuring the reliability of predicting renewable energy production with a reasonable prediction horizon of 3–24 hours. With this framework, rMDC operators can reliably execute virtual machine placement and migration to minimize the impact of renewable energy variability. The SkyBox framework in particular has the ability to reduce the carbon footprint of rMDCs by nearly 46%. Another such WMF is the SHIELD framework (Qi et al., 2024), which creates a scheduling policy for workloads based on workload and node type. Through this framework, energy costs, carbon emissions, and water usage were able to be reduced significantly; up to 1.3, 3.7, and 1.8 times the reductions of other WMFs respectively. This framework optimizes a variety of sustainability metrics (such as PUE, WUE, and CUE), while also considering the economic factors of data centers.

When it comes to measuring energy usage in software, there are generally two kinds of approaches. The first approach estimates hardware energy consumption during software execution by utilizing specialized software tools, such as PowerAPI or Joulemeter (Verón et al., 2024). These tools derive estimates from a model fed with parameters of the computing environment collected during runtime. Although this approach is easy to implement and can give useful results, their reliance on estimation implies that the accuracy depends purely on the approximations made by the model. The second approach to this is directly recording energy consumption during software execution through the usage of measurement equipment on a suitably instrumented computer (Verón et al., 2024). This method provides genuine energy consumption data, assuming that the equipment and measurement quality are up to standard. The issues with this approach, however, is that it requires a costly and arduous implementation and computer instrumentation for proper testing. Both of these methods, though, offer software

developers the ability to identify energy hotspots, diagnose energy consumption, and help optimize energy consumption to work towards more sustainable software (Noureddine, 2024).

Future Actions

Looking at what can be done from here, the first step should be for governments to implement some of these recommendations into future policies. Incentivize the use of hardware that aligns with federally-recommended sustainability guidelines, such as Energy Star or EPEAT. Additionally, incentivize or even mandate sustainable practices within data centers on power and hardware management, such as automating utilization management, power management, and optimizing cooling fan/pump speeds. Lastly, mandate the measurement of usage effectiveness and other useful metrics for analyzing energy usage in data centers. Doing so would allow for a better understanding of data center energy use and its drivers, which could be of aid when designing and evaluating effective policies, especially as “data center energy use enters a critical transition phase” (Masanet et al., 2020). The economics, environmental factors, and energy consumption within ICT are all closely related with each other (Rodionov et al., 2022) and must be considered for sustainable development, especially when developing policies for the ICT sector.

Another effective way to bolster a greener, more sustainable ICT industry is to invest in new potential technologies in computing, data storage, communications, and heat removal. This would help in making these technologies economically viable and scalable in time to prevent a demand surge, especially once current efficiency trends reach their limits. One such example is the quantum computer, a device that manipulates information using the mathematics of quantum mechanics and has seen a lot of potential for future implementation, though energy usage is still undetermined (Chen, 2023). A modern technology with a growing focus on its potential is cooling systems for data centers. From using free air cooling (Qi et al., 2024) to using liquid and immersion cooling technologies (Masanet et al., 2020), heat removal technologies have the potential to dramatically reduce ICT energy usage and eliminate compressor based cooling. One example of this comes from an air-cooled data center located in Singapore, which observed approximately 37% cooling energy savings when the temperature target increased from 20°C to 32°C (Van Le & Tan, 2024). With this aspect of data centers, there is also the potential to capture

heat directly into the cooling system and recycle it for other purposes such as space heating, which can help in achieving a PUE less than 1 (Gelenbe & Caseau, 2015). There are many more serious options that exist today, such as further chip specialization and ultrahigh density storage to name a few, but these make up only a fraction of the potential innovations for ICT.

References

- Chen, S. (2023, June 26). *Are quantum computers really energy efficient?* Nat Comput Sci 3, 457-460. <https://doi.org/10.1038/s43588-023-00459-6>
- Masanet, E., Shehabi, A., Lei, N., Smith, S., Koomey, J. (2020, February 28). *Recalibrating global data center energy-use estimates*. Science, 267 (6481), 3. <https://www-science-org.ezp1.lib.umn.edu/doi/10.1126/science.aba3758>
- Sun, J., Gong, Z., Agarwal, A., Noghabi, S., Chandra, R., Snir, M., Huang, J. (2024, November 20). *Exploring the Efficiency of Renewable Energy-based Modular Data Centers at Scale*. <https://dl-acm-org.ezp2.lib.umn.edu/doi/10.1145/3698038.3698544>
- Qi, S., Milojevic, D., Bash, C., & Pasricha, S. (2024, May 29). *SHIELD: Sustainable Hybrid Evolutionary Learning Framework for Carbon, Wastewater, and Energy-Aware Data Center Management*. <https://dl-acm-org.ezp2.lib.umn.edu/doi/10.1145/3634769.3634810>
- Van Le, D., & Tan, R. (2024, May 31). *An Optimization Approach for Selection and Deployment of Energy-Efficient Servers in Green Data Centers*. <https://dl-acm-org.ezp3.lib.umn.edu/doi/10.1145/3632775.3662829>
- Noureddine, A. (2024, May 23). *Analyzing Software Energy Consumption*. <https://dl-acm-org.ezp2.lib.umn.edu/doi/10.1145/3639478.3643058>
- Gelenbe, E., & Caseau, Y. (2015, June 29). *The impact of information technology on energy consumption and carbon emissions*. Ubiquity, 2015 (June), 15. <https://dl-acm-org.ezp2.lib.umn.edu/doi/10.1145/2755977>
- Rodionov, D., Daniali, S., Khortabi, F., & Moqaddasnejad, A. (2022, September 28). *Study Effect of Information and Communication Technology (ICT) on Energy Consumption and Greenhouse Gas Emissions in Selected Oil-producing Countries*. <https://dl-acm-org.ezp2.lib.umn.edu/doi/10.1145/3527049.3527123>

Energy Act of 2020 IT Sustainability Best Practice Guide. (2023, March).

https://www.cio.gov/assets/resources/Energy%20Act%20of%202020%20IT%20Best%20Practices%20Guide_FINAL_v2.0.pdf

Verón, J., Pérez, C., Calero, C., Moraga, M., Pérez, F., & Cetina, C. (2024, September 22). A Comparative Analysis of Energy Consumption Between Visual Scripting models and C++ in Unreal Engine: Raising Awareness on the importance of Green MDD.

<https://dl-acm-org.ezp2.lib.umn.edu/doi/10.1145/3640310.3674099>